Basic Hadoop Command Structure

Most Hadoop commands follow this structure:

hadoop [subcommand] [options] [arguments]

Example:

hadoop fs -ls /

This lists files in the **HDFS root directory**.

HDFS File System Commands (hadoop fs)

These commands work like **Linux file commands**, but they operate on **HDFS (Hadoop Distributed File System)**.

Check Hadoop Version

hadoop version

Shows installed Hadoop version.

List Files

hadoop fs -ls /

Example output:

drwxr-xr-x - user supergroup 0 2024-01-10 /input

Create Directory

hadoop fs -mkdir /data

Create nested directories:

hadoop fs -mkdir -p /data/logs

Upload File from Windows to HDFS

hadoop fs -put C:\data\file.txt /input

or

hadoop fs -copyFromLocal C:\data\file.txt /input

Download File from HDFS to Windows

hadoop fs -get /input/file.txt C:\output

or

hadoop fs -copyToLocal /input/file.txt C:\output

Display File Content

hadoop fs -cat /input/file.txt

Remove File

hadoop fs -rm /input/file.txt

Delete directory:

hadoop fs -rm -r /input

Check Disk Usage

hadoop fs -du /input

Check File Size

hadoop fs -stat %b /input/file.txt

Hadoop Distributed File System Commands (hdfs dfs)

Modern Hadoop uses **hdfs dfs** instead of hadoop fs.

Example:

```
hdfs dfs -ls /
```

Common commands:

```
hdfs dfs -mkdir /data
hdfs dfs -put file.txt /data
hdfs dfs -get /data/file.txt
hdfs dfs -rm /data/file.txt
```

MapReduce Commands

Run MapReduce jobs.

Example:

```
hadoop jar hadoop-mapreduce-examples.jar wordcount /input /output
```

Explanation:

Part	Meaning
hadoop	Hadoop command
jar	Run a jar program
examples.jar	MapReduce program
wordcount	Job name
/input	Input folder
/output	Output folder

Hadoop Daemon Commands (Windows)

Start Hadoop services.

Start NameNode

hdfs namenode

Start DataNode

hdfs datanode

Start ResourceManager

yarn resourcemanager

Start NodeManager

yarn nodemanager

Check HDFS Health

hdfs fsck /

Checks:

- corrupted blocks
- missing files
- replication status

Hadoop Admin Commands

Safe Mode

Enter safe mode:

hdfs dfsadmin -safemode enter

Leave safe mode:

hdfs dfsadmin -safemode leave

Report Cluster Status

hdfs dfsadmin -report

Shows:

- number of DataNodes
- storage usage
- cluster health

YARN Commands

Check cluster nodes:

yarn node -list

List applications:

yarn application -list

Kill application:

yarn application -kill <app_id>

Hadoop Web Interfaces

After starting Hadoop:

Service	URL
NameNode	http://localhost:9870
ResourceManager	http://localhost:8088

Example Full Workflow

Step 1 Create folder

hdfs dfs -mkdir /input

Step 2 Upload file

hdfs dfs -put C:\data\sample.txt /input

Step 3 Run WordCount

hadoop jar hadoop-mapreduce-examples.jar wordcount /input /output

Step 4 View result

hdfs dfs -cat /output/part-r-00000

Common Hadoop Commands Cheat Sheet

Command	Purpose
hadoop version	check version
hdfs dfs -ls	list files
hdfs dfs -mkdir	create directory
hdfs dfs -put	upload file
hdfs dfs -get	download file
hdfs dfs -rm	delete file
hdfs dfs -cat	view file
hadoop jar	run MapReduce job